

Supplementary Material

RASM: Relationship-Aware Augmentation Selection through Structural Redundancy Modeling

Prasad Kanhegaonkar, Surya Prakash

Department of Computer Science and Engineering
Indian Institute of Technology Indore, India

July 16, 2026

This supplementary material provides methodological details and experimental results that complement the main manuscript. It includes descriptions of the candidate augmentation pool, class-aware sample selection strategy, preliminary correlation measure analysis, selected representative augmentation subsets, threshold sensitivity analysis, and qualitative comparisons of augmentation strategies. These additional analyses provide implementation details and supporting evidence that could not be included in the main manuscript because of space limitations, further demonstrating the effectiveness and robustness of the proposed Relationship-Aware Augmentation Selection Method (RASM) across different dermoscopic datasets.

1 Candidate Augmentation Pool

The proposed RASM framework constructs an initial candidate augmentation pool consisting of 54 image transformations implemented using the Albumentations image augmentation library. The augmentation pool includes a diverse set of geometric, color and intensity, noise, blur, channel manipulation, deformation, and weather-related transformations that are commonly employed in image classification tasks. Rather than applying all augmentations during training, RASM analyzes the similarity relationships among these candidate transformations to identify structurally redundant augmentations and select a compact subset of representative operations. Figure 1 provides a visual overview of the candidate augmentation pool used throughout the study.

2 Class-Aware Sample Selection

	HAM10000 Dataset							ISIC2019 Dataset							
Class	AKIEC	BCC	BKL	DF	MEL	NV	VASC	AKIEC	BCC	BKL	DF	MEL	NV	SCC	VASC
Classwise Image Count	327	514	1099	115	1113	6705	142	867	3323	2624	239	4522	12875	628	253
Per Class Frequency	0.033	0.051	0.110	0.012	0.111	0.670	0.014	0.034	0.131	0.104	0.009	0.179	0.508	0.025	0.010
Negative Log	1.486	1.290	0.960	1.940	0.954	0.174	1.848	1.466	0.882	0.985	2.025	0.748	0.294	1.606	2.000
Max-Min Normalization	0.743	0.632	0.445	1.0	0.442	0.0	0.948	0.677	0.340	0.399	1.0	0.263	0.0	0.758	0.986
Scaled Max-Min Values	16.145	14.476	11.672	20.0	11.626	5.0	19.222	15.152	10.096	10.985	20.0	8.937	5.0	16.365	19.786
Selected Images Count	53	74	128	23	129	335	27	131	335	288	48	404	644	103	50

Table 1: Class-aware sample selection strategy used by RASM to construct the representative dataset for augmentation relationship modeling.

To ensure that the augmentation relationship modeling stage is computationally efficient while remaining representative of the original datasets, a class-aware sampling strategy is employed to construct a balanced subset of training images. Rather than selecting an equal number of samples from each class, the proposed strategy allocates samples according to the inverse frequency of each class. Specifically, the class frequencies are first computed from the original dataset, followed by a negative logarithmic

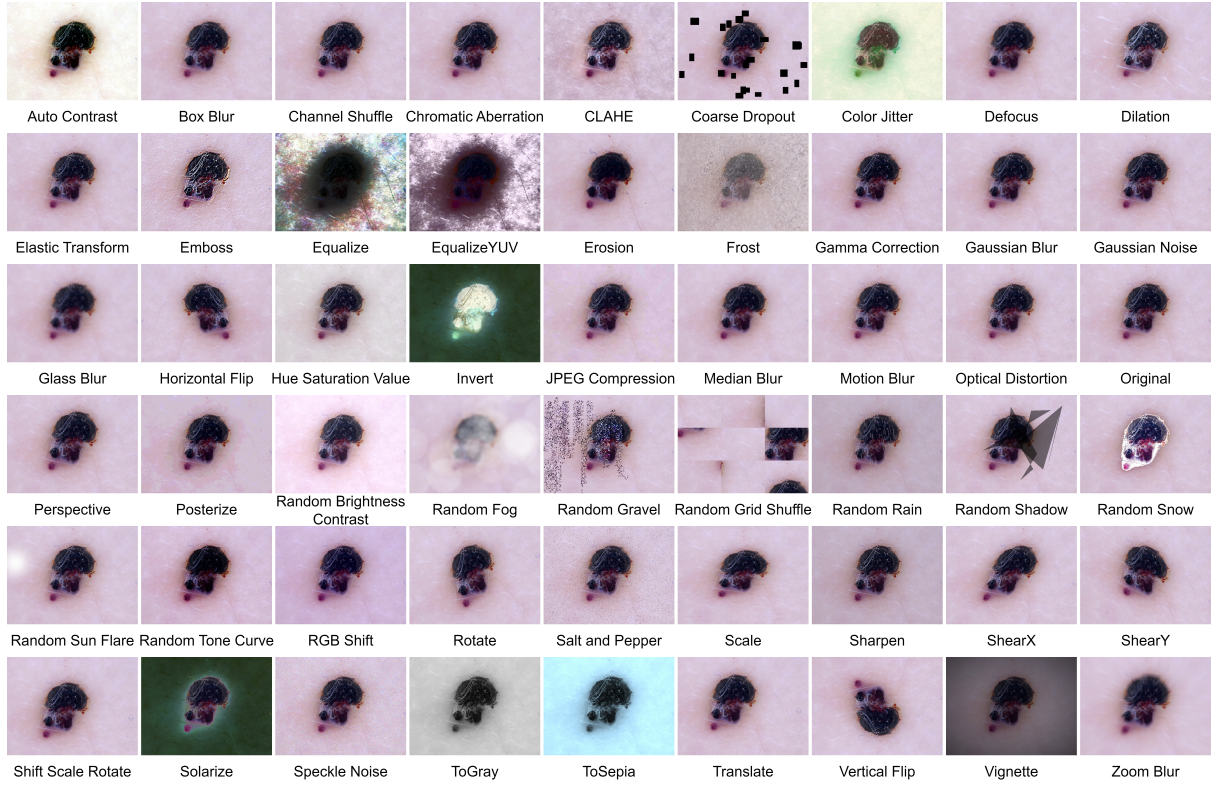


Figure 1: Candidate augmentation operations used to construct the augmentation pool in RASM.

transformation to reduce the dominance of majority classes while preserving their relative proportions. The transformed values are subsequently normalized using min-max normalization and linearly scaled to determine the desired sampling weights. The final number of selected images for each class is obtained by multiplying these weights with the corresponding class frequencies and rounding to the nearest integer.

Table 1 summarizes each stage of the sampling procedure for the HAM10000 and ISIC2019 datasets, including the original class distributions, normalized sampling weights, and the final number of representative images selected from each class. This representative subset is subsequently used to generate augmentation pairs for learning the similarity relationships among candidate augmentation transformations.

3 Limitations of Traditional Augmentation Selection Policies

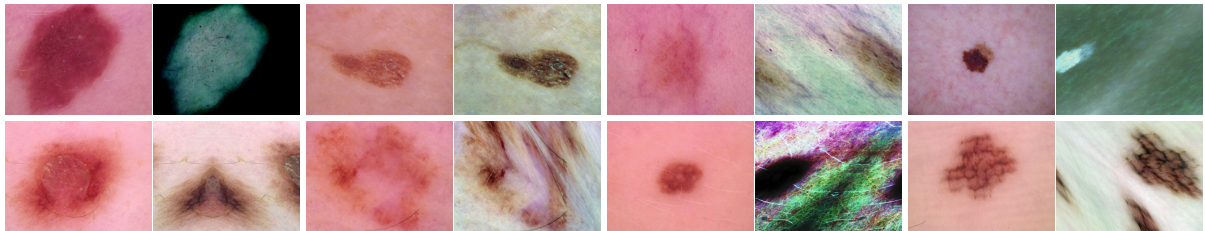


Figure 2: Representative examples of RandAugment transformations applied to dermoscopic images. Some randomly sampled transformation combinations substantially alter lesion appearance or introduce artifacts that are less representative of clinically plausible variations.

Figure 2 illustrates representative examples of images generated using RandAugment on dermoscopic images. Since RandAugment randomly samples augmentation operations without considering the semantic relationships among transformations or the characteristics of the target domain, certain transformation combinations may substantially alter lesion appearance or introduce artifacts that are less represen-

tative of clinically plausible image variations. Such transformations can reduce the semantic consistency between the original and augmented images, potentially affecting the quality of the learned feature representations. This limitation highlights the importance of selecting augmentation operations based on their semantic relationships rather than relying solely on random policy sampling. The proposed RASM framework addresses this issue by identifying representative augmentations through relationship-aware structural redundancy modeling, thereby preserving augmentation diversity while avoiding redundant or unsuitable transformations.

4 Candidate Augmentation Descriptions

	Augmentation	Description	Parameter & Value
Geometric	Rotate	Rotate image by <i>mag</i> degrees.	Angle=[-45,45]
	Translate	Translate image by <i>mag</i> percent.	Magnitude=[-0.1,0.1]
	Scale	Scale image by <i>magnitude</i> percent.	Magnitude=[0.7,1.3]
	SSR	Shift, scale & rotate together.	
	Flip	Random horizontal or vertical flip.	
	Shear	Random shear in X or Y direction.	Degrees=[-20,20]
Color and Intensity	ColorJitter	Change brightness, contrast, saturation by <i>x</i> and hue by <i>y</i> .	<i>x</i> =[0.8,1.2], <i>y</i> =[-0.5,0.5]
	Bright/Cont	Change brightness/contrast.	Limit=[-0.2,0.2]
	Hue Sat Value	Change hue, saturation, and value.	Limit=[-20,20]
	Sharpen	Enhancing edge details.	α =[0.2,0.5], L =[0.5,1.0]
	Gamma	Simulate different light conditions.	Gamma Limit=[80,120]
	Invert	Subtracting pixel values from 255.	
	Posterize	Reduce number of bits per channel.	Num Bits=[4]
	Solarize	Invert pixel values above threshold.	Threshold=[128]
	Auto Contrast	Dynamically adjust image contrast.	Cutoff Percentile=[3]
	Chr. Aberration	Distorting red and blue channels.	Mode=Random
	RGB Shift	Shift values for RGB channels.	Limit=[-20,20]
	ToGray	Convert image to grayscale.	Method=Weighted Avg
	ToSepia	Apply sepia filter to RGB image.	
Noise	Gaussian	Apply Gaussian noise to input image.	Variance Range=[10,50]
	Salt & Pepper	Randomly set pixels to black/white.	Probability=[0.01]
	Speckle	Multiplicative noise added to pixels.	Probability=[0.1]
	JPEG	JPEG compression-induced noise.	Quality Range=[0.1,0.3]
Blur	Box	Apply box blur.	Kernel Size=[3,5]
	Gaussian	Apply Gaussian kernel to blur image.	Kernel Size=[3,7]
	Motion	Simulate motion-induced blurring.	Kernel Size=[3,7]
	Median	Apply median blur to image.	Kernel Size=[3,7]
	Glass	Apply a glass blur effect to image.	σ =[0.7], δ =[4], Iter=[2]
	Zoom	Apply zoom blur transform.	Max Factor=[1,1.31]
	Defocus	Apply defocus blur to image.	Radius=[3,5]
Channel	Emboss	Apply emboss effect to image.	Strength=[1.0]
	Channel Shuffle	Randomly rearrange RGB channels.	
	CLAHE	Apply CLAHE algorithm.	
	Equalize	Equalize histogram in RGB space.	
Deformations	Equalize_yuv	Equalize histogram in YUV space.	
	Elastic	Stretch/compress in elastic manner.	α =3, σ =50
	Optical	Apply optical distortion.	Limit=[0.2]
	Perspective	Simulating changes in camera angle.	Scale=[0.2]
	Grid Shuffle	Rearranging patches within image.	Grid size=[3,3]
	Coarse Dropout	Masking random square regions.	Num Holes=[10,20]
	Morphological	Erosion/Dilation transformations.	Scale=[3,5]
Weather Transforms	Vignette	Adding vignette effect to image.	Intensity=[0.6]
	Shadow	Simulating shadows in the image.	Num Shadows=[2,3]
	Rain	Add rain effects in image.	Bright=0.9, W=1, Blur=3
	Snow	Bleach out pixels imitating snow.	Bright=2, Snow=[0.4,0.7]
	Sun Flare	Simulates Sun Flare in image.	R=150, Angle=0.8
	Fog	Simulates fog in image.	Coef=[0.5,0.8], α =0.2
	Gravel	Add gravel in image.	ROI=(0,0,1,1), Patch=40
	Tone Curve	Alter relationship bw bright/dark.	Scale=[0.2]
Weather Transforms	Frost	Apply frost effect to image.	α =[0.5]

Table 2: Data augmentation methods considered in the proposed work

The proposed RASM framework considers a diverse pool of 54 candidate augmentation operations implemented using the Albumentations library. These transformations are organized into seven categories: geometric, color and intensity, noise, blur, channel manipulation, deformations, and weather transforms. Each augmentation is associated with a predefined parameter range to ensure realistic variations while preserving the semantic characteristics of dermoscopic images. During the relationship modeling stage, these candidate augmentations are evaluated to identify structurally redundant transformations, from which a compact subset of representative augmentations is automatically selected. Table 2 summarizes

the augmentation operations together with their corresponding descriptions and parameter settings used throughout this study.

5 Threshold Sensitivity Analysis

Threshold	HAM10000		ISIC2019	
	Num.	Aug.	Num.	Aug.
0.05	2	0.6757	2	0.6320
0.10	2	0.6705	2	0.6486
0.15	2	0.7293	3	0.5942
0.20	3	0.6753	4	0.6275
0.25	4	0.6800	7	0.6214
0.30	5	0.7302	8	0.6336
0.35	7	0.6612	14	0.6603
0.40	9	0.6622	16	0.6451
0.45	15	0.7087	26	0.6575
0.50	17	0.7079	40	0.6869
0.55	23	0.7130	48	0.6708
0.60	39	0.7293	51	0.6580
0.65	47	0.7472	54	0.6324
0.70	51	0.6713	54	0.6656
0.75	54	0.6918	54	0.6532
0.80	54	0.7353	54	0.6687
0.85	54	0.7090	54	0.6573
0.90	54	0.7070	54	0.6756
0.95	54	0.7261	54	0.6164

Table 3: Complete threshold sensitivity analysis used to determine the optimal similarity threshold.

The similarity threshold governs the construction of the augmentation relationship graph and directly influences the number of representative augmentations selected by RASM. Table 3 reports the complete threshold sensitivity analysis for the HAM10000 and ISIC2019 datasets, including the number of selected augmentations and the corresponding F1-score obtained at each threshold. The optimal thresholds are selected as 0.65 for HAM10000 and 0.50 for ISIC2019, as they achieve the highest downstream F1-score while maintaining compact representative transformation subsets.

6 Preliminary Comparison of Correlation Measures

Method	TH	AUG	ACC	F1	AUC	SEN	SPEC	PRE	JI
Spearman	0.20	3	95.90	95.89	98.87	91.88	99.78	95.93	92.29
	0.25	5	96.35	96.38	99.50	94.25	99.65	96.46	93.18
	0.30	10	96.60	96.60	99.51	95.13	99.73	96.63	93.54
	0.35	25	96.45	96.50	99.73	95.43	99.74	96.64	93.35
	0.40	39	96.40	96.45	99.79	96.04	99.65	96.57	93.28
	0.45	46	96.35	96.39	99.69	96.27	99.66	96.49	93.13
	0.50	49	96.35	96.36	99.68	92.98	99.78	96.43	93.14
	0.55	51	96.25	96.30	99.64	94.77	99.72	96.43	93.01
	0.60	52	96.35	96.34	99.18	94.34	99.79	96.37	93.07
Kendall	0.15	4	96.30	96.29	99.29	91.33	99.76	96.30	93.00
	0.20	7	96.50	96.50	99.52	93.28	99.76	96.53	93.39
	0.25	32	96.40	96.44	99.72	93.98	99.77	96.54	93.27
	0.30	44	96.50	96.49	99.41	94.09	99.81	96.50	93.37
	0.45	53	92.45	92.50	98.71	86.09	99.60	92.64	86.53

Table 4: Preliminary comparison of Spearman’s ρ and Kendall’s τ for augmentation relationship modeling on the HAM10000 dataset. Only threshold values producing distinct augmentation subsets are reported. Thresholds yielding identical augmentation subsets or selecting all candidate augmentations are omitted for brevity.

During the development of the proposed RASM framework, preliminary experiments were conducted to compare Spearman’s ρ and Kendall’s τ as measures of augmentation similarity. For each correlation measure, augmentation relationship graphs were constructed using different similarity thresholds, and the resulting representative augmentation subsets were evaluated through downstream classification performance.

Table 4 summarizes the experimental results on the HAM10000 dataset. Only threshold values that produced distinct augmentation subsets are reported. Thresholds yielding identical connected components and, consequently, identical augmentation subsets are omitted because they do not provide additional information. Furthermore, threshold values that resulted in the selection of all 54 candidate augmentations are excluded, as they are equivalent to the standard *All Aug* baseline and therefore do not contribute to the comparison between Spearman’s ρ and Kendall’s τ .

The results show that both correlation measures achieve comparable downstream classification performance. However, Kendall’s τ produces fewer distinct augmentation subsets across different threshold values, indicating a more stable and less redundant grouping of augmentation transformations. Consequently, Kendall’s τ was adopted as the similarity measure in the proposed RASM framework, while Spearman’s ρ was not considered in the subsequent experiments presented in the main manuscript.

7 Implementation Details

The proposed framework was implemented in Python using TensorFlow 2.15. Image augmentations were generated using Albumentations library. Graph construction was performed using adjacency matrices, while connected components were identified using depth-first search (DFS). The Siamese network was initialized with ImageNet-pretrained EfficientNet-B0 weights and trained using the Adam optimizer with the hyperparameters reported in the main manuscript. Classification experiments used the same EfficientNet-B0 backbone and identical training settings to ensure fair comparison among augmentation strategies. All experiments were conducted on an NVIDIA Tesla A100 GPU.

8 Selected Representative Transformations

Category	Selected Transformations
Geometric	Rotate, Translate, Scale, ShiftScaleRotate, HorizontalFlip, VerticalFlip, ShearX, ShearY
Color & Intensity	ColorJitter, RandomBrightnessContrast, HueSaturationValue, Sharpen, Gamma, Invert, Posterize, Solarize, AutoContrast, ChromaticAberration, RGBShift, ToGray
Noise	GaussNoise, SaltandPepper, Speckle, JPEGCompression
Blur	BoxBlur, MotionBlur, ZoomBlur
Channel	ChannelShuffle, CLAHE, Equalize, EqualizeYUV
Deformations	ElasticTransform, OpticalDistortion, Perspective, RandomGridShuffle, CoarseDropout, Erosion, Dilation, Vignette
Weather Transforms	RandomRain, RandomSnow, RandomSunFlare, RandomFog, RandomGravel, RandomToneCurve, Frost
Other	Original

Table 5: Representative transformation subset selected by RASM for the HAM10000 dataset at the optimal similarity threshold ($\tau = 0.65$). The reported subset consists of 47 selected transformations, including the original image.

Category	Selected Transformations
Geometric	Rotate, Translate, Scale, ShiftScaleRotate, HorizontalFlip, VerticalFlip, ShearX, ShearY
Color & Intensity	AutoContrast, ColorJitter, HueSaturationValue, RandomBrightnessContrast, Invert, Posterize, Solarize, RGBShift, ToGray, Frost
Noise	JPEGCompression
Blur	BoxBlur, GlassBlur, Emboss, ZoomBlur
Channel	ChannelShuffle, CLAHE, EqualizeYUV
Deformations	Perspective, OpticalDistortion, RandomGridShuffle, CoarseDropout, Dilation, Vignette
Weather	RandomFog, RandomGravel, RandomRain, RandomShadow, RandomSnow, RandomSunFlare, RandomToneCurve
Other	Original

Table 6: Representative transformation subset selected by RASM for the ISIC2019 dataset at the optimal similarity threshold ($\tau = 0.50$). The reported subset consists of 40 selected transformations, including the original image.

The proposed RASM framework selected 47 representative transformations for the HAM10000 dataset and 40 representative transformations for the ISIC2019 dataset at their respective optimal similarity thresholds. Each selected subset includes the original image together with the representative augmentation operations.

The representative transformation subsets differ between the two datasets, demonstrating that RASM automatically adapts the selected transformations according to dataset-specific augmentation relationships learned during the relationship modeling stage. While several transformation categories are retained across both datasets, the composition and size of the final subsets vary, reflecting differences in the underlying augmentation similarity structure learned for each dataset.

9 Qualitative Comparison of Augmentation Strategies



Figure 3: Qualitative comparison of augmentation strategies on the HAM10000 and ISIC2019 datasets. Compared with applying the complete augmentation pool, the representative transformation subsets selected by the proposed RASM framework preserve clinically meaningful appearance variations while avoiding unrealistic transformations, synthetic artifacts, and semantically inconsistent image modifications.

Figure 3 compares representative images obtained using three augmentation strategies: the original images, the complete augmentation pool, and the representative transformation subsets selected by the proposed RASM framework. Across both datasets, applying all augmentation operations occasionally produces unrealistic color distortions, artificial artifacts, and clinically implausible appearances. In contrast, the representative transformations selected by RASM better preserve lesion morphology and visual characteristics while introducing meaningful appearance variations, resulting in more semantically consistent augmented samples.